ABC MANUFACTURING CORPORATION

Environmental Health & Safety Department

STANDARD OPERATING PROCEDURE

Chemical Spill Response

Document No. SOP-EHS-003

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A. Document Control

A.1 Revision History

Rev	Date	Description of Change	Author	Approver
00	19 Mar 2019	Initial issue of the chemical spill response procedure.	J. Alvarez	M. Okafor
01	14 Jun 2023	Added incidental vs. emergency classification and HAZWOPER response tiers; added drain-protection steps.	R. Nguyen	M. Okafor
02	15 Jan 2026	Biennial review. Added risk matrix, spill decision tree, absorbent compatibility guide, and RQ reporting note.	J. Alvarez	M. Okafor

A.2 Distribution List

Copy No.	Holder / Role	Location	Format
Master	Document Owner — Manager, EHS	DMS (electronic master)	Electronic
01	Production Floor Supervisors	Line 1–4 supervisor stations	Controlled print
02	Emergency Response Team Lead	ERT muster point	Controlled print
03	Chemical Store / Receiving Lead	Chemical storage building	Controlled print
04	Maintenance Planner	Maintenance office	Electronic
05	Training Coordinator	Learning Management System	Electronic

A.3 Controlled Copy Statement

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1. Purpose

This Standard Operating Procedure establishes the requirements for the safe, prompt, and effective response to chemical spills and releases at ABC Manufacturing Corporation. Its purpose is to protect personnel, the environment, and property; to define the boundary between an incidental spill that trained employees may clean up and an emergency release that requires the Emergency Response Team (ERT); and to ensure releases are contained, cleaned up, disposed of, and reported in accordance with applicable regulations.

- Ensure every spill is assessed before any response and correctly classified as incidental or emergency.
- Prevent released chemicals from reaching personnel, floor drains, soil, or waterways.
- Ensure only trained personnel with adequate controls attempt cleanup, and that emergencies trigger evacuation and the ERT.
- Support the site ISO 14001 environmental and ISO 45001 OH&S management systems and regulatory release-reporting obligations.

2. Scope

This procedure applies to all ABC Manufacturing employees, contractors, and temporary workers at any ABC Manufacturing facility where hazardous chemicals are present.

2.1 Inclusions

- Assessment and classification of spills, incidental spill cleanup, containment, decontamination, waste handling, notification, and documentation.
- Initial defensive actions (evacuation, isolation, notification) prior to ERT arrival for emergency releases.

2.2 Exclusions

- Offensive emergency response entry, plugging, and patching, which are performed only by HAZWOPER-qualified responders under the Incident Commander and the site emergency response plan.
- Uncontrolled bulk tank or process releases (site emergency plan and Process Safety program).
- Non-chemical spills (e.g., water, non-hazardous materials) and hazardous-waste accumulation, addressed by the site waste procedure.

3. References

This procedure is based on publicly available regulations and consensus standards. Where a conflict exists, the most stringent applicable requirement governs.

Reference	Title / Subject
29 CFR 1910.120	Hazardous waste operations and emergency response (HAZWOPER)
29 CFR 1910.120(q)	Emergency response to hazardous substance releases
29 CFR 1910.38	Emergency action plans
29 CFR 1910.1200	Hazard Communication — SDS Section 6, accidental release measures
29 CFR 1910.157	Portable fire extinguishers
NFPA 30 / NFPA 704	Flammable and Combustible Liquids; hazard identification
ANSI/ISEA Z358.1	Emergency Eyewash and Shower Equipment
40 CFR 112 / EPCRA	Spill Prevention (SPCC) for oil; emergency release reporting (as applicable)
ISO 14001:2015 / ISO 45001:2018	Environmental and OH&S management systems

Reference	Title / Subject
Internal	SOP-EHS-001 Chemical Handling & Storage; SOP-EHS-002 HazCom; SOP-EHS-013 Incident Reporting; SOP-EHS-015 Root Cause Analysis

4. Definitions

Term	Definition
Incidental Spill	A release of limited quantity and low hazard that can be safely absorbed, neutralized, or otherwise controlled by employees in the immediate area with available equipment, without evacuation.
Emergency Release	A release that poses (or could pose) a significant safety or health hazard, requires evacuation, involves a highly toxic or uncontrolled material, or exceeds the capability of area employees — requiring the ERT under HAZWOPER.
HAZWOPER	OSHA's hazardous waste operations and emergency response standard, defining responder tiers (Awareness, Operations, Technician, Specialist, Incident Commander).
IDLH	Immediately Dangerous to Life or Health — an atmosphere posing an immediate threat to life or that would impair escape.
Reportable Quantity (RQ)	A regulatory threshold above which a release of a listed substance must be reported to designated agencies.
Absorbent	An inert or reactive material used to soak up or neutralize a spilled liquid (universal, oil-only, or acid/base neutralizing).
Decontamination	The process of removing or neutralizing chemical contamination from personnel, PPE, and equipment.
Spill Kit	A staged assembly of absorbents, PPE, containment, and disposal materials matched to the chemicals in the area.

5. Responsibilities

Role	Responsibility
First Discoverer (any employee)	Raise the alarm, protect self and others, evacuate if unsafe, and notify the supervisor. Does not attempt cleanup beyond training.
Area Supervisor	Classifies the spill with EHS input, directs incidental cleanup by trained staff, initiates emergency response and evacuation as needed.
Emergency Response Team (ERT)	Performs emergency response entry, source control, and containment under the Incident Commander per HAZWOPER.
Incident Commander (IC)	Assumes command of an emergency release, manages responders, and authorizes offensive actions and re-entry.
EHS	Supports classification, arranges waste disposal, determines reporting obligations, and leads the investigation.
Maintenance	Maintains spill equipment, drains, containment, and eyewash/shower infrastructure.

6. Required Training

Response duties are limited to the responder's HAZWOPER training tier. No employee performs a task beyond their level of training.

Training Module	Audience	Frequency
HAZWOPER First Responder Awareness	All employees near chemicals	Initial + annual
HAZWOPER First Responder Operations (defensive)	ERT members	Initial (8 hr) + annual refresher
Spill kit use, containment & drain protection	Trained cleanup staff, ERT	Initial + 2-yr refresh

Training Module	Audience	Frequency
Chemical Spill Response (this SOP)	All chemical handlers, supervisors	Initial + on revision
Decontamination & waste handling	ERT, cleanup staff	Initial + 2-yr refresh

7. Required Qualifications

- Awareness-level training for any employee who may discover a release; Operations-level for anyone performing defensive containment.
- Medical clearance and fit-testing for any responder required to wear respiratory protection.
- Demonstrated competency in spill kit deployment and drain protection during drills, signed off by a Competent Person.

8. Required PPE

PPE for incidental cleanup is selected from SDS Section 8 and the chemical involved. Emergency response PPE (including chemical-protective suits and respiratory protection) is specified by the Incident Commander and the emergency response plan.

Response Task	Eye/Face	Hand	Body	Respiratory
Incidental spill — low-hazard liquid	Goggles	Chemical-resistant gloves	Coverall / apron	Per SDS
Incidental spill — corrosive	Goggles + face shield	Chemical-resistant gauntlet	Chemical apron + chemical boots	Per SDS
Defensive containment (Operations)	Goggles / face shield	Chemical-resistant gloves	Chemical suit as required	APR/PAPR per assessment
Emergency entry	Per IC	Per IC	Per IC (up to encapsulating)	SAR/SCBA per IC

CAUTION	Never enter an atmosphere that is or may be IDLH without SCBA/SAR and IC authorization. If in doubt about the atmosphere, evacuate and treat as an emergency.

9. Required Equipment

- Spill kits matched to area chemicals: universal (general), oil-only, and acid/base neutralizing kits, inspected monthly and after each use.
- Absorbents, socks/booms, and pillows; neutralizers with pH indicators; non-sparking scoops and tools where flammables are present.
- Drain covers/plugs and dike/berm materials to protect floor drains and prevent environmental release.
- Overpack drums and labeled hazardous-waste containers for spent absorbent and residue.
- ANSI/ISEA Z358.1 eyewash/drench shower, Class B extinguisher, gas/vapor detection where required, and area communication.

10. Hazard Identification

- **Chemical exposure** — skin/eye contact, inhalation of vapors, or ingestion during response.
- **Fire / reactivity** — ignition of flammable vapors, or heat/gas from reactive or incompatible materials.
- **Oxygen-deficient / IDLH atmosphere** — vapor accumulation in low or confined areas.
- **Environmental release** — spill reaching drains, soil, or waterways triggering reporting obligations.

- **Slips / falls** — wet or slippery surfaces from the spill or from response water.
- **Incompatible cleanup** — using the wrong absorbent/neutralizer causing heat or gas generation.

11. Risk Assessment

Risk level is derived from the severity × likelihood matrix. Any release assessed as High or Critical is handled as an emergency and escalated to the ERT before any cleanup is attempted.

11.1 Risk Rating Matrix

Severity ↓ / Likelihood →	Rare	Unlikely	Possible	Likely	Almost Certain
Catastrophic	Medium	High	High	Critical	Critical
Major	Medium	Medium	High	High	Critical
Moderate	Low	Medium	Medium	High	High
Minor	Low	Low	Medium	Medium	High
Negligible	Low	Low	Low	Low	Medium

11.2 Task Risk Register

Hazard	Severity	Likelihood	Risk Level	Control Measures
Inhalation of vapor during cleanup	Major	Possible	**High**	Assess atmosphere, ventilate, respiratory protection per SDS, evacuate if IDLH possible.
Flammable spill ignites	Catastrophic	Unlikely	**High**	Remove ignition sources, non-sparking tools, Class B extinguisher, treat large spill as emergency.
Corrosive splash / burn to responder	Major	Possible	**High**	Goggles/face shield, gauntlets, chemical apron/boots, Z358.1 eyewash within 10 s.
Spill reaches floor drain / environment	Moderate	Possible	**Medium**	Deploy drain covers first, dike/berm, spill kit, notify EHS for reporting review.
Wrong absorbent causes reaction	Moderate	Unlikely	**Medium**	Use compatible absorbent/neutralizer (App. C), check SDS §10 before applying.
Slip/fall on wet surface	Minor	Likely	**Medium**	Barricade area, signage, absorb promptly, non-slip footwear.

12. Detailed Step-by-Step Procedure

12.1 Pre-Job Preparation (Readiness)

1. Verify spill kits are stocked, sealed, and matched to area chemicals; confirm drain covers, PPE, eyewash/shower, and extinguisher are serviceable.
2. Know the location of the nearest spill kit, alarm, eyewash/shower, exits, and the SDS for chemicals in the area.

12.2 Spill Discovery & Assessment

1. On discovering a spill, do not rush in. From a safe distance, identify the chemical from the label/SDS and estimate the quantity.
2. Assess hazards: flammability, toxicity, reactivity, vapor, proximity to drains/personnel, and whether the atmosphere may be IDLH.
3. Classify the spill using the decision tree in Appendix A. If any emergency criterion applies, go directly to Section 13.

WARNING	If the material is unknown, highly toxic, generating vapor, on fire, large, or spreading to drains — treat it as an EMERGENCY. Evacuate, alarm, and summon the ERT. Do not attempt cleanup.

12.3 Immediate Actions (Incidental Spill)

1. Alert others in the immediate area and restrict access; post a spill/wet-floor barricade.
2. Eliminate ignition sources for flammable materials; do not operate switches that could spark.
3. Don PPE per Section 8 and the SDS before approaching the spill.

12.4 Source Control & Containment

1. Stop the source only if it is safe and within training — right a tipped container, close a valve, or apply a temporary plug.
2. Protect drains FIRST using covers/plugs; then dike or surround the spill with socks/booms to stop spreading.
3. Work from the outside of the spill inward to prevent enlarging the affected area.

12.5 Cleanup & Decontamination

1. Apply the compatible absorbent or neutralizer for the chemical (Appendix C); for acids/bases, neutralize and verify with pH indicator before pickup.
2. Collect saturated absorbent and residue into a labeled hazardous-waste/overpack container; do not mix incompatible waste.
3. Decontaminate tools, PPE, and the affected surface; bag disposable PPE as contaminated waste.

12.6 Verification

1. Confirm the area is clean and free of residue, drains are clear and uncontaminated, and no vapor remains.
2. Confirm all waste is containerized, labeled, and staged for disposal, and that the spill kit is restocked.

12.7 Housekeeping & Waste Disposal

1. Arrange hazardous-waste disposal through EHS; complete the waste label and log. Never pour spill residue down a drain.
2. Restock the spill kit, replace used PPE, and return the area to service once cleared by the supervisor.

12.8 Documentation

1. Complete the Spill Report (Appendix D) and, for any injury/exposure or environmental release, an incident report per SOP-EHS-013.
2. EHS reviews the event for regulatory reporting (RQ) and initiates root cause analysis (SOP-EHS-015) where required.

12.9 Escalation Procedures

1. Spill grows, vapor increases, or a responder feels symptoms → stop, withdraw, and escalate to emergency response.
2. Release reaches a drain, soil, or waterway → notify EHS immediately for reporting determination.
3. Any doubt about classification → default to emergency and summon the ERT.

13. Emergency Response

WARNING	Emergency releases are handled by trained ERT responders under the Incident Commander only. All other personnel evacuate and account for at the muster point.

13.1 Immediate Actions

- Activate the alarm, evacuate personnel upwind/uphill, and isolate the area. Do not re-enter.

- Eliminate ignition sources en route out for flammable releases; assist injured persons only if it is safe.

13.2 Medical Response

- For exposure, flush at the eyewash/drench shower for at least 15 minutes and seek medical aid; provide the SDS to responders and clinicians.

13.3 Fire Response

- If a flammable spill ignites beyond incipient stage, evacuate and let the fire brigade/ERT respond; use SDS Section 5 media only for incipient fires if trained.

13.4 Source Control & Containment (ERT only)

- The ERT performs source control, containment, and confinement under the IC using the emergency response plan and appropriate PPE.

13.5 Equipment Shutdown

- Where safe, isolate energy and process feeds to the release from outside the hazard area; do not re-enter to shut down.

13.6 Notification Chain

Step	Notify	Contact / Method	When
1	Area Supervisor	Radio / floor phone	Immediately on any release
2	EHS On-Call	Ext. 2400 / site radio	Within 5 minutes
3	Emergency Response Team / IC	Emergency line 2222	Any emergency release, fire, or injury
4	External Emergency Services	911	Life threat, fire, or release beyond site control
5	EHS Manager & Plant Manager	Per call tree	Within 30 minutes; reportable/RQ events
6	Regulatory agencies (via EHS)	Per reporting matrix	When an RQ or reportable threshold is exceeded

14. Required Documentation

Record	Purpose	Owner	Retention
Spill Report (App. D)	Capture spill details, response, and disposal	Area Supervisor / EHS	5 years
Spill Kit Inspection Log (App. B)	Monthly readiness of spill equipment	Area Supervisor	3 years
Hazardous Waste Label / Manifest	Track disposal of spill residue	EHS	Per regulation
Incident / Exposure Report	Injuries, exposures, environmental release	EHS	5 years
Training Records (LMS)	Evidence of qualified responders	Training Coordinator	Duration + 3 years
Regulatory Notification Record	Evidence of RQ/agency reporting	EHS	Per regulation

15. References (Public Standards)

- OSHA 29 CFR 1910.120 — HAZWOPER, including 1910.120(q) emergency response.
- OSHA 29 CFR 1910.38 — Emergency action plans.
- OSHA 29 CFR 1910.1200 — Hazard Communication (SDS Section 6).
- OSHA 29 CFR 1910.157 — Portable fire extinguishers.

- NFPA 30; NFPA 704 — Flammable liquids; hazard identification.
- ANSI/ISEA Z358.1 — Emergency Eyewash and Shower Equipment.
- EPA 40 CFR 112 (SPCC) and EPCRA release reporting, as applicable.
- ISO 14001:2015; ISO 45001:2018 — Management-system standards.

16. Appendices

Appendix A — Spill Classification Decision Tree

> START → Is anyone injured, is there fire, or is the atmosphere possibly IDLH? — YES → EMERGENCY: evacuate, alarm, ERT (Section 13).
>
> NO → Is the material unknown, highly toxic, or generating significant vapor? — YES → EMERGENCY.
>
> NO → Is the spill large, spreading, or reaching a drain/soil/waterway beyond your control? — YES → EMERGENCY (protect drains if safe en route out).
>
> NO → Are trained personnel, correct PPE, and a compatible spill kit available? — NO → Isolate and escalate to supervisor/ERT.
>
> YES → INCIDENTAL SPILL: don PPE, protect drains, contain, clean up, decontaminate, document (Section 12) → END.

Appendix B — Spill Kit Monthly Inspection Checklist

#	Item	OK	Action Needed
1	Kit sealed, present at assigned location, and unobstructed	☐	☐
2	Absorbents / socks / pillows stocked to par level	☐	☐
3	Neutralizer + pH indicator present and in date	☐	☐
4	PPE (gloves, goggles/face shield, apron, boots) present	☐	☐
5	Drain covers / plugs present	☐	☐
6	Overpack / waste bags and labels present	☐	☐
7	Contents match area chemical hazards	☐	☐
8	Kit restocked after last use (if applicable)	☐	☐

Inspected by: ________________________ Date: ____________ Location / Kit ID: ____________

Appendix C — Absorbent / Neutralizer Compatibility Guide

Spilled Material	Recommended Absorbent / Neutralizer	Avoid
Flammable/combustible liquids	Universal or oil-only absorbent; non-sparking tools	Reactive/acid neutralizers; ignition sources
Acids	Acid-neutralizing absorbent (indicating); neutralize before pickup	Combustible organic absorbents on strong oxidizing acids
Bases (caustics)	Base-neutralizing absorbent (indicating); neutralize before pickup	Acid neutralizer on large volumes without control
Oxidizers	Inert/inorganic absorbent (e.g., vermiculite)	Organic/combustible absorbent (fire risk)
Oil / hydrocarbons	Oil-only (hydrophobic) absorbent, booms	—
Unknown	Do not apply absorbent — treat as emergency	Any absorbent until identified

Always confirm compatibility against SDS Sections 7 and 10 before applying any absorbent or neutralizer.

Appendix D — Chemical Spill Report Form

Field	Entry
Report No. / Date / Time	
Location / Area	
Chemical & approximate quantity	
Classification	☐ Incidental ☐ Emergency
Reached drain / soil / waterway?	☐ No ☐ Yes (describe)
Source / cause	
Response actions taken	
PPE used	
Injuries / exposures	☐ None ☐ Yes (describe; ref. incident report)
Waste generated / disposal method	
Notifications made (who / when)	
Regulatory reporting required?	☐ No ☐ Yes (agency / time)
Supervisor signature / date	
EHS follow-up (RCA ref. SOP-EHS-015)	